DATA 200 - LAB1

CheckMyGrade Application

Github link: https://github.com/rutujasaste06/CheckMyGrade

Implementation:

The application starts with a main menu where users can register or log in. After a successful login, users are shown different options based on their role (admin, professor, or student). Each role has its own functions and tasks. All the data is saved and managed using CSV files.

Students.csv

Courses.csv

Professors.csv

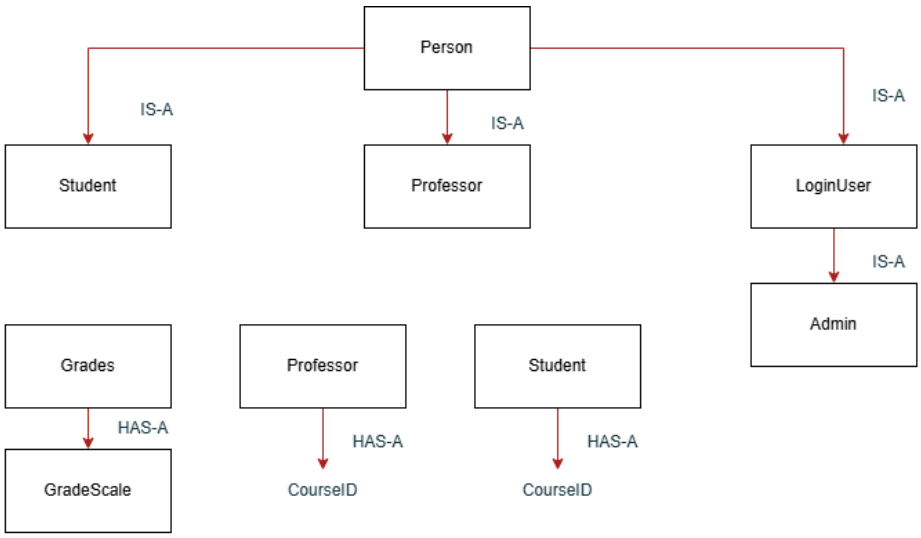
Login.csv

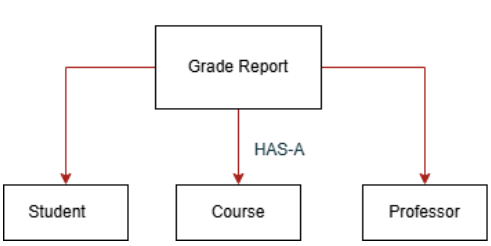
Classes used:

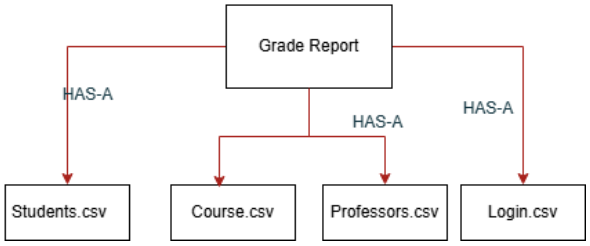
- CSVFileHandler – Handles all file operations for CSV files (read,write,update,delete)
- Person – Base class for student,professor and Loginuser which stores common attributes as name and email
- Student – Represents student records and stores name,email,course,grade,marks and provides functionalities as add,delete,update student records (calling csvfilehandler), view grades, search by email or course.
- Professor – Represents professor records by storing name,email,rank and course and provides functionalities add,delete,update professor details and view assigned courses
- Course – Represents course information by storing coursed,name,description and provides add,delete,update functionalities
- LoginUser – Handles user authentication and login system by storing email, encrypted password,role and provides register user,login/logout,change password functionalities.
- Admin – This inherits loginUser class and provides functionalities as manage users,delete accounts,reset passwords,change roles,view system summary
- Grades – Manages grading system by storing grade ranges(A,B,C,etc) and provides conversion of marks to grades and add/delete/modify grade levels.

- Searchsort – Handles searching and sorting operations which includes binary search, linear search,sorting,quick sort implementation. This measures execution time.

IS/HAS-A relationship:







1. Case1: Main screen of the application

```
═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  1. Login
  2. Register
  0. Exit
  Select: 1
```

2. Register screen of the application for admin,student and professor

```
TERMINAL    PROBLEMS    OUTPUT    DEBUG CONSOLE    PORTS

  ── Register New User ──────────────────────────────
  Email (unique ID): moham.M@sjsu.edu
  Password: Moham*
  Role (student / professor / admin): Professor
  Full name: Mohammad Masum
  Rank: Senior Professor
  Course ID they teach: DATA220
  User 'moham.M@sjsu.edu' registered successfully.

  [Press Enter to continue]
```

```
═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  1. Login
  2. Register
  0. Exit
  Select: 2

  ── Register New User ──────────────────────────────
  Email (unique ID): tom.c@sjsu.edu
  Password: TomC*
  Role (student / professor / admin): student
  First name: Tom
  Last name: C
  Course ID to enroll in: DATA226
  Initial marks (0-100, or Enter for 0): 0\
  Error: could not convert string to float: '0\\'

  [Press Enter to continue]
```

3. Login screen for admin:

```
TERMINAL    PROBLEMS    OUTPUT    DEBUG CONSOLE    PORTS

  — Login ————————————————————————————————————
  Email: admin1@sjsu.edu
  — Login ————————————————————————————————————
  Email: admin1@sjsu.edu
  Password: Admin1*

  Password: Admin1*

  Welcome, admin1@sjsu.edu! Role: admin

═══════════════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════════════
  Logged in as: admin1@sjsu.edu  [admin]

  1. Student Management
  2. Course Management
  3. Professor Management
  4. User Accounts
  5. Grade Reports & Statistics
  6. Grade Scale Settings
  7. Change Password
  0. Logout
  Select: 1
```

- Admin screen: Student management: Display student records

```
TERMINAL    PROBLEMS    OUTPUT    DEBUG CONSOLE    PORTS

═══════════════════════════════════════════════════════════════
  Logged in as: admin1@sjsu.edu  [admin]

  1. Student Management
  2. Course Management
  3. Professor Management
  4. User Accounts
  5. Grade Reports & Statistics
  6. Grade Scale Settings
  7. Change Password
  0. Logout
  Select: 1

  — Student Management  [Admin] ———————————————
  1. Display all student records
  2. Add student enrollment
  3. Delete all enrollments for a student
  4. Drop one course for a student
  5. Update marks/grade
  6. Search by email
  7. Search by name
  8. Search by course
  9. Sort by marks (QuickSort)
  A. Sort by name
  B. Sort by email
  0. Back
  Select: 1

  Email                    Name              Course     Grade   Marks
  ---------------------------------------------------------------------
  john.jacobs@sjsu.edu     John Jacobs       DATA200    F         0.0
  marc.j@sjsu.edu          Marc J            DATA200    B        86.0
  john1.jacobs@sjsu.edu    John Jacobs       DATA200    B        86.0
  mike.k@sjsu.edu          Mike K            DATA200    A        90.0
  rutu.s@sjsu.edu          Rutu S            DATA226    F         0.0
  rutu.s@sjsu.edu          Rutu S            DATA200    A        96.0
  mike.k@sjsu.edu          Mike K            DATA220    B        85.0
  john1.jacobs@sjsu.edu    John Jacobs       DATA220    A        95.0

  [Press Enter to continue]

  — Student Management  [Admin] ———————————————
  1. Display all student records
  2. Add student enrollment
```

- Admin screen: Student management: Add student enrollment of registered students

```
═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  1. Login
  2. Register
  0. Exit
  Select: 2

  ── Register New User ──────────────────────────────
  Email (unique ID): kevin.j@sjsu.edu
  Password: Kevinj*
  Role (student / professor / admin): student
  First name: Kevin
  Last name: J
  Course ID to enroll in: DATA220
  Initial marks (0-100, or Enter for 0): 0
  User 'kevin.j@sjsu.edu' registered successfully.
```

```
Logged in as: admin1@sjsu.edu  [admin]

1. Student Management
2. Course Management
3. Professor Management
4. User Accounts
5. Grade Reports & Statistics
6. Grade Scale Settings
7. Change Password
0. Logout
Select: 1

── Student Management  [Admin] ──────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 2
First name: Kevin
Last name: J
Email: kevin.j@sjsu.edu
Course ID: DATA220
Marks (0-100): 0
Error: Student 'kevin.j@sjsu.edu' is already enrolled in 'DATA220'.

[Press Enter to continue]
```

- Admin screen: Student management: Drop one course for a student

```
Email                     Name              Course     Grade   Marks
---------------------------------------------------------------------------
marc.j@sjsu.edu           Marc J            DATA200    B        86.0
john1.jacobs@sjsu.edu     John Jacobs       DATA200    B        86.0
mike.k@sjsu.edu           Mike K            DATA200    A        90.0
rutu.s@sjsu.edu           Rutu S            DATA226    B        89.0
rutu.s@sjsu.edu           Rutu S            DATA200    A        96.0
mike.k@sjsu.edu           Mike K            DATA220    B        85.0
john1.jacobs@sjsu.edu     John Jacobs       DATA220    A        95.0
tom.c@sjsu.edu            Tom C             DATA226    A        92.0
tom.c@sjsu.edu            Tom C             DATA220    B        89.0
kevin.j@sjsu.edu          Kevin J           DATA220    F         0.0

[Press Enter to continue]

── Student Management  [Admin] ──────────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 4
Student email: mike.k@sjsu.edu
Course ID: DATA200
Dropped.
```

```
Email                     Name              Course     Grade   Marks
---------------------------------------------------------------------------
marc.j@sjsu.edu           Marc J            DATA200    B        86.0
john1.jacobs@sjsu.edu     John Jacobs       DATA200    B        86.0
rutu.s@sjsu.edu           Rutu S            DATA226    B        89.0
rutu.s@sjsu.edu           Rutu S            DATA200    A        96.0
mike.k@sjsu.edu           Mike K            DATA220    B        85.0
john1.jacobs@sjsu.edu     John Jacobs       DATA220    A        95.0
tom.c@sjsu.edu            Tom C             DATA226    A        92.0
tom.c@sjsu.edu            Tom C             DATA220    B        89.0
kevin.j@sjsu.edu          Kevin J           DATA220    F         0.0
```

- Admin screen: Student management: Update one course for a student

```
Email                    Name             Course     Grade   Marks
---------------------------------------------------------------------
marc.j@sjsu.edu          Marc J           DATA200    B        86.0
john1.jacobs@sjsu.edu    John Jacobs      DATA200    B        86.0
rutu.s@sjsu.edu          Rutu S           DATA226    B        89.0
rutu.s@sjsu.edu          Rutu S           DATA200    A        96.0
mike.k@sjsu.edu          Mike K           DATA220    B        85.0
john1.jacobs@sjsu.edu    John Jacobs      DATA220    A        95.0
tom.c@sjsu.edu           Tom C            DATA226    A        92.0
tom.c@sjsu.edu           Tom C            DATA220    B        89.0
kevin.j@sjsu.edu         Kevin J          DATA220    F         0.0

[Press Enter to continue]

── Student Management  [Admin] ──────────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 5
Student email: kevin.j@sjsu.edu
Course ID: DATA220
New marks (blank to keep): 88
New first name (blank to keep):
New last name (blank to keep):
Updated.
```

- Admin screen: Student management: Search by email

```
── Student Management  [Admin] ──────────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 6
Email to search: rutu.s@sjsu.edu

Time for Binary search by email: 0.0139 ms  (1 record(s) found)

Grades for Rutu S (rutu.s@sjsu.edu)
Course       Grade     Marks
------------------------------
DATA226      B          89.0
DATA200      A          96.0

[Press Enter to continue]
```

- Admin screen: Student management: Search by name

```
— Student Management  [Admin] ———————————————
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 7
Name substring: Rutu

Time for Linear search by name: 0.0064 ms  (2 record(s) found)

Email                     Name              Course     Grade   Marks
---------------------------------------------------------------------
rutu.s@sjsu.edu           Rutu S            DATA226    B        89.0
rutu.s@sjsu.edu           Rutu S            DATA200    A        96.0
```

- Admin screen: Student management: Search by course

```
— Student Management  [Admin] ———————————————
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 8
Course ID: DATA220

Time for Linear search by course: 0.0054 ms  (4 record(s) found)

Email                     Name              Course     Grade   Marks
---------------------------------------------------------------------
mike.k@sjsu.edu           Mike K            DATA220    B        85.0
john1.jacobs@sjsu.edu     John Jacobs       DATA220    A        95.0
tom.c@sjsu.edu            Tom C             DATA220    B        89.0
kevin.j@sjsu.edu          Kevin J           DATA220    B        88.0

[Press Enter to continue]
```

- Admin screen: Student management: Sort by marks

```
── Student Management  [Admin] ──────────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: 9

Time for QuickSort by marks: 9.3799 ms

Email                     Name              Course     Grade   Marks
-----------------------------------------------------------------------------
mike.k@sjsu.edu           Mike K            DATA220    B        85.0
marc.j@sjsu.edu           Marc J            DATA200    B        86.0
john1.jacobs@sjsu.edu     John Jacobs       DATA200    B        86.0
kevin.j@sjsu.edu          Kevin J           DATA220    B        88.0
tom.c@sjsu.edu            Tom C             DATA220    B        89.0
rutu.s@sjsu.edu           Rutu S            DATA226    B        89.0
tom.c@sjsu.edu            Tom C             DATA226    A        92.0
john1.jacobs@sjsu.edu     John Jacobs       DATA220    A        95.0
rutu.s@sjsu.edu           Rutu S            DATA200    A        96.0

[Press Enter to continue]
```

- Admin screen: Student management: Sort by name

```
── Student Management  [Admin] ──────────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: A

Time for Sort by name: 0.0039 ms

Email                     Name              Course     Grade   Marks
-----------------------------------------------------------------------
john1.jacobs@sjsu.edu     John Jacobs       DATA200    B        86.0
john1.jacobs@sjsu.edu     John Jacobs       DATA220    A        95.0
kevin.j@sjsu.edu          Kevin J           DATA220    B        88.0
marc.j@sjsu.edu           Marc J            DATA200    B        86.0
mike.k@sjsu.edu           Mike K            DATA220    B        85.0
rutu.s@sjsu.edu           Rutu S            DATA226    B        89.0
rutu.s@sjsu.edu           Rutu S            DATA200    A        96.0
tom.c@sjsu.edu            Tom C             DATA226    A        92.0
tom.c@sjsu.edu            Tom C             DATA220    B        89.0

[Press Enter to continue]
```

- Admin screen: Student management: Sort by email

```
── Student Management  [Admin] ──────────────────────
1. Display all student records
2. Add student enrollment
3. Delete all enrollments for a student
4. Drop one course for a student
5. Update marks/grade
6. Search by email
7. Search by name
8. Search by course
9. Sort by marks (QuickSort)
A. Sort by name
B. Sort by email
0. Back
Select: B

Time for Sort by email: 0.0049 ms

Email                     Name              Course     Grade   Marks
---------------------------------------------------------------------
john1.jacobs@sjsu.edu     John Jacobs       DATA200    B        86.0
john1.jacobs@sjsu.edu     John Jacobs       DATA220    A        95.0
kevin.j@sjsu.edu          Kevin J           DATA220    B        88.0
marc.j@sjsu.edu           Marc J            DATA200    B        86.0
mike.k@sjsu.edu           Mike K            DATA220    B        85.0
rutu.s@sjsu.edu           Rutu S            DATA226    B        89.0
rutu.s@sjsu.edu           Rutu S            DATA200    A        96.0
tom.c@sjsu.edu            Tom C             DATA226    A        92.0
tom.c@sjsu.edu            Tom C             DATA220    B        89.0
```

- Admin screen: Course management: Display all courses

```
═══════════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════════
  Logged in as: admin1@sjsu.edu  [admin]

  1. Student Management
  2. Course Management
  3. Professor Management
  4. User Accounts
  5. Grade Reports & Statistics
  6. Grade Scale Settings
  7. Change Password
  0. Logout
  Select: 2

  ── Course Management  [Admin] ──────────────────────
  1. Display all courses
  2. Add course
  3. Delete course
  4. Modify course
  5. Search by name
  0. Back
  Select: 1

  ID          Name                    Description
  ------------------------------------------------------------
  DATA200     Computer Programming    Python programming
  DATA226     Data warehousing        Data
  DATA220     Math methods for DI     Maths

  [Press Enter to continue]
```

- Admin screen: Course management: Add course

```
── Course Management  [Admin] ──────────────────────
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 2
Course ID: DATA201
Course name: SQL
Description: SQL
Added.
```

- Admin screen: Course management: Delete course

```
── Course Management  [Admin] ──────────────────────
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 1

ID          Name                    Description
------------------------------------------------------------
DATA200     Computer Programming    Python programming
DATA226     Data warehousing        Data
DATA220     Math methods for DI     Maths
DATA201     SQL                     SQL

[Press Enter to continue]

── Course Management  [Admin] ──────────────────────
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 3
Course ID to delete: DATA201
Deleted.
```

- Admin screen: Course management: Modify course

```
DATA200      Computer Programming      Python programming
DATA226      Data warehousing          Data
DATA220      Math methods for DI       Maths
DATA201      SQL                       SQL

[Press Enter to continue]

── Course Management  [Admin] ──────────────────────
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 3
Course ID to delete: DATA201
Deleted.

[Press Enter to continue]

── Course Management  [Admin] ──────────────────────
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 4
Course ID to modify: DATA226
New course name: DATA warehouse
New description: Data warehousing and functionalities
Updated.

[Press Enter to continue]

── Course Management  [Admin] ──────────────────────
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 1

ID           Name                      Description
------------------------------------------------------------
DATA200      Computer Programming      Python programming
DATA226      DATA warehouse            Data warehousing and functionalities
DATA220      Math methods for DI       Maths

[Press Enter to continue]
```

- Admin screen: Course management: Search by name

```
— Course Management  [Admin] ————————————————————
1. Display all courses
2. Add course
3. Delete course
4. Modify course
5. Search by name
0. Back
Select: 5
Course name substring: math

Time for Search courses by name: 0.0093 ms  (1 record(s) found)

ID          Name                      Description
------------------------------------------------------------
DATA220     Math methods for DI       Maths

[Press Enter to continue]
```

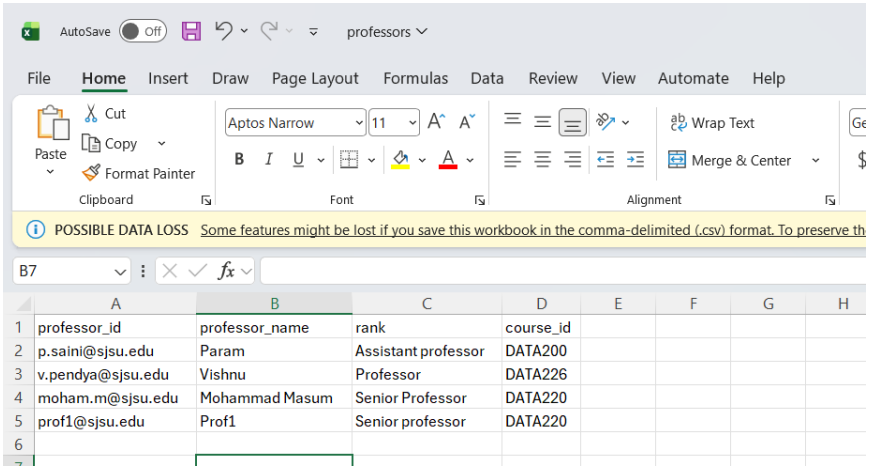
- Admin screen: Professor management: Display all professors

```
═══════════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════════
 Logged in as: admin1@sjsu.edu  [admin]

 1. Student Management
 2. Course Management
 3. Professor Management
 4. User Accounts
 5. Grade Reports & Statistics
 6. Grade Scale Settings
 7. Change Password
 0. Logout
 Select: 3

 — Professor Management  [Admin] ——————————————
 1. Display all professors
 2. Add professor
 3. Delete professor
 4. Modify professor
 5. Search by name
 0. Back
 Select: 1

 Email                    Name              Rank                 Course
 ---------------------------------------------------------------------------
 p.saini@sjsu.edu         Param             Assistant professor  DATA200
 v.pendya@sjsu.edu        Vishnu            Professor            DATA226
 moham.m@sjsu.edu         Mohammad Masum    Senior Professor     DATA220

 [Press Enter to continue]
```

- Admin screen: Professor management: Add professors

```
— Professor Management  [Admin] ———————————
1. Display all professors
2. Add professor
3. Delete professor
4. Modify professor
5. Search by name
0. Back
Select: 2
Full name: Prof1
Email: prof1@sjsu.edu
Rank: Senior professor
Course ID: DATA220
Added.

[Press Enter to continue]

— Professor Management  [Admin] ———————————
1. Display all professors
2. Add professor
3. Delete professor
4. Modify professor
5. Search by name
0. Back
Select: 1

Email                    Name              Rank                 Course
-------------------------------------------------------------------------------
p.saini@sjsu.edu         Param             Assistant professor  DATA200
v.pendya@sjsu.edu        Vishnu            Professor            DATA226
moham.m@sjsu.edu         Mohammad Masum    Senior Professor     DATA220
prof1@sjsu.edu           Prof1             Senior professor     DATA220
```

- Added in profesors.csv file

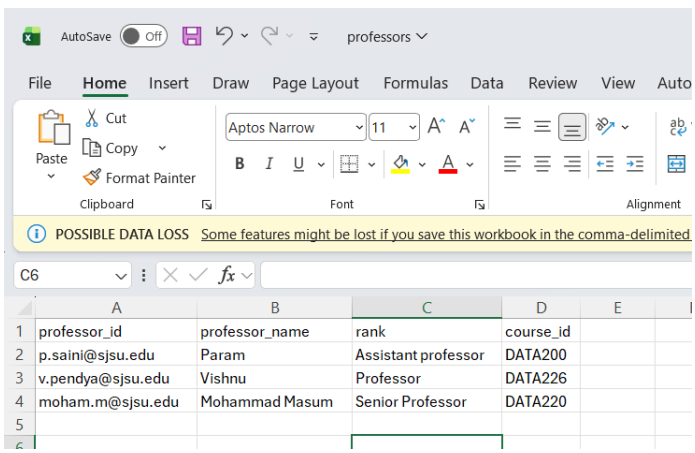


professor_id	professor_name	rank	course_id
p.saini@sjsu.edu	Param	Assistant professor	DATA200
v.pendya@sjsu.edu	Vishnu	Professor	DATA226
moham.m@sjsu.edu	Mohammad Masum	Senior Professor	DATA220
prof1@sjsu.edu	Prof1	Senior professor	DATA220

- Admin screen: Professor management: Delete professors

```
[Press Enter to continue]

— Professor Management  [Admin] ———————————
1. Display all professors
2. Add professor
3. Delete professor
4. Modify professor
5. Search by name
0. Back
Select: 3
Professor email to delete: prof1@sjsu.edu
Deleted.

[Press Enter to continue]

— Professor Management  [Admin] ———————————
1. Display all professors
2. Add professor
3. Delete professor
4. Modify professor
5. Search by name
0. Back
Select: 1

Email                     Name              Rank                 Course
--------------------------------------------------------------------------------
p.saini@sjsu.edu          Param             Assistant professor  DATA200
v.pendya@sjsu.edu         Vishnu            Professor            DATA226
moham.m@sjsu.edu          Mohammad Masum    Senior Professor     DATA220

[Press Enter to continue]
```

- Updated in professors.csv file



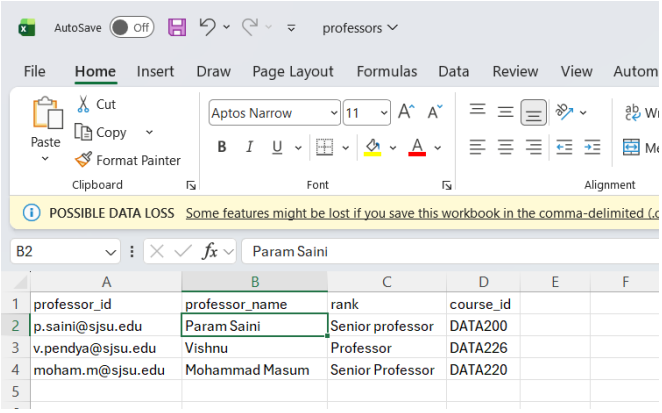
	A	B	C	D
1	professor_id	professor_name	rank	course_id
2	p.saini@sjsu.edu	Param	Assistant professor	DATA200
3	v.pendya@sjsu.edu	Vishnu	Professor	DATA226
4	moham.m@sjsu.edu	Mohammad Masum	Senior Professor	DATA220
5				

- Admin screen: Professor management: Modify professors

```
[Press Enter to continue]

— Professor Management  [Admin] ———————————————
1. Display all professors
2. Add professor
3. Delete professor
4. Modify professor
5. Search by name
0. Back
Select: 4
Professor email to modify: p.saini@sjsu.edu
New name: Param Saini
New rank: Senior professor
New course ID: DATA200
Updated.

[Press Enter to continue]
```

```
Email                    Name              Rank              Course
-------------------------------------------------------------------------------
p.saini@sjsu.edu         Param Saini       Senior professor  DATA200
v.pendya@sjsu.edu        Vishnu            Professor         DATA226
moham.m@sjsu.edu         Mohammad Masum    Senior Professor  DATA220
```

- Updated in professors.csv file



	A	B	C	D	E	F
1	professor_id	professor_name	rank	course_id		
2	p.saini@sjsu.edu	Param Saini	Senior professor	DATA200		
3	v.pendya@sjsu.edu	Vishnu	Professor	DATA226		
4	moham.m@sjsu.edu	Mohammad Masum	Senior Professor	DATA220		
5						

- Admin screen: Professor management: Search by name

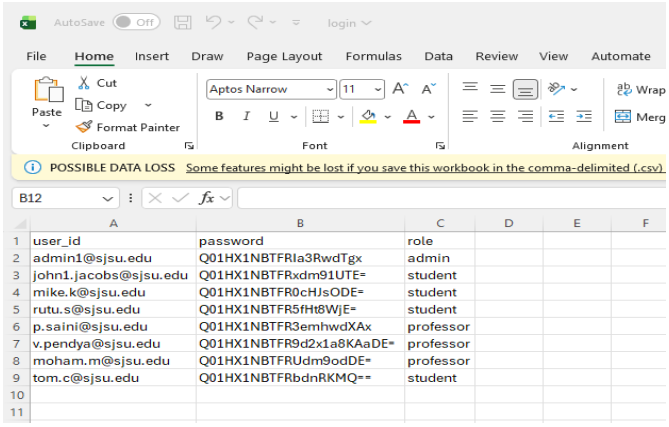
```
— Professor Management  [Admin] ———————————————
1. Display all professors
2. Add professor
3. Delete professor
4. Modify professor
5. Search by name
0. Back
Select: 5
Name substring: Vishnu

Time for Search professors by name: 0.0163 ms  (1 record(s) found)

Email                    Name              Rank              Course
-------------------------------------------------------------------------------
v.pendya@sjsu.edu        Vishnu            Professor         DATA226

[Press Enter to continue]
```

- Admin screen: User accounts: View all

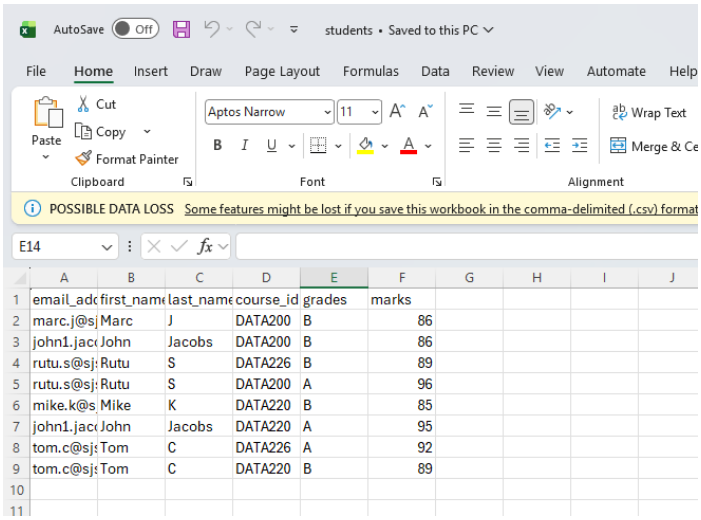
```
═══════════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════════
 Logged in as: admin1@sjsu.edu  [admin]

 1. Student Management
 2. Course Management
 3. Professor Management
 4. User Accounts
 5. Grade Reports & Statistics
 6. Grade Scale Settings
 7. Change Password
 0. Logout
 Select: 4

 — User Accounts  [Admin] ————————————————————
 1. View all accounts
 2. Delete a user account
 3. Reset a user's password
 4. Change a user's role
 5. System summary
 0. Back
 Select: 1

 Email                        Role        Password (encrypted)
 ------------------------------------------------------------------------------
 admin1@sjsu.edu              admin       Q01HX1NBTFRIa3RwdTgx...
 john1.jacobs@sjsu.edu        student     Q01HX1NBTFRxdm91UTE=...
 mike.k@sjsu.edu              student     Q01HX1NBTFR0cHJsODE=...
 rutu.s@sjsu.edu              student     Q01HX1NBTFR5fHt8WjE=...
 p.saini@sjsu.edu             professor   Q01HX1NBTFR3emhwdXAx...
 v.pendya@sjsu.edu            professor   Q01HX1NBTFR9d2x1a8KAaDE=...
 moham.m@sjsu.edu             professor   Q01HX1NBTFRUdm9odDE=...
 tom.c@sjsu.edu               student     Q01HX1NBTFRbdnRKMQ==...
 kevin.j@sjsu.edu             student     Q01HX1NBTFRSbH1wdXEx...

 [Press Enter to continue]
```

- Admin screen: User accounts: Delete a user account – This will delete the user account from student/professor and login.csv files

```
 — User Accounts  [Admin] ————————————————————
 1. View all accounts
 2. Delete a user account
 3. Reset a user's password
 4. Change a user's role
 5. System summary
 0. Back
 Select: 1

 Email                        Role        Password (encrypted)
 ------------------------------------------------------------------------------
 admin1@sjsu.edu              admin       Q01HX1NBTFRIa3RwdTgx...
 john1.jacobs@sjsu.edu        student     Q01HX1NBTFRxdm91UTE=...
 mike.k@sjsu.edu              student     Q01HX1NBTFR0cHJsODE=...
 rutu.s@sjsu.edu              student     Q01HX1NBTFR5fHt8WjE=...
 p.saini@sjsu.edu             professor   Q01HX1NBTFR3emhwdXAx...
 v.pendya@sjsu.edu            professor   Q01HX1NBTFR9d2x1a8KAaDE=...
 moham.m@sjsu.edu             professor   Q01HX1NBTFRUdm9odDE=...
 tom.c@sjsu.edu               student     Q01HX1NBTFRbdnRKMQ==...
 kevin.j@sjsu.edu             student     Q01HX1NBTFRSbH1wdXEx...

 [Press Enter to continue]

 — User Accounts  [Admin] ————————————————————
 1. View all accounts
 2. Delete a user account
 3. Reset a user's password
 4. Change a user's role
 5. System summary
 0. Back
 Select: 2
 Email of user to delete: kevin.j@sjsu.edu
 Removed 2 related row(s).

 [Press Enter to continue]
```

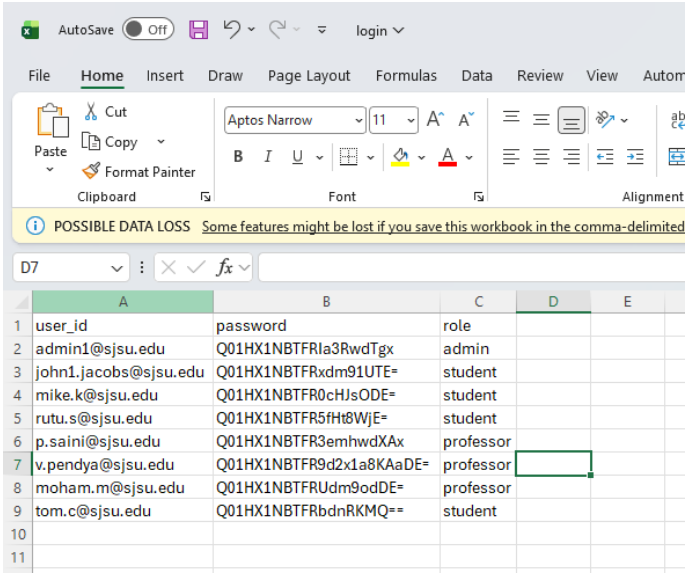


	A	B	C
1	user_id	password	role
2	admin1@sjsu.edu	Q01HX1NBTFRIa3RwdTgx	admin
3	john1.jacobs@sjsu.edu	Q01HX1NBTFRxdm91UTE=	student
4	mike.k@sjsu.edu	Q01HX1NBTFR0cHJsODE=	student
5	rutu.s@sjsu.edu	Q01HX1NBTFR5fHt8WjE=	student
6	p.saini@sjsu.edu	Q01HX1NBTFR3emhwdXAx	professor
7	v.pendya@sjsu.edu	Q01HX1NBTFR9d2x1a8KAaDE=	professor
8	moham.m@sjsu.edu	Q01HX1NBTFRUdm9odDE=	professor
9	tom.c@sjsu.edu	Q01HX1NBTFRbdnRKMQ==	student



	A	B	C	D	E	F
1	email_adc	first_name	last_name	course_id	grades	marks
2	marc.j@sj	Marc	J	DATA200	B	86
3	john1.jac	John	Jacobs	DATA200	B	86
4	rutu.s@sj	Rutu	S	DATA226	B	89
5	rutu.s@sj	Rutu	S	DATA200	A	96
6	mike.k@s	Mike	K	DATA220	B	85
7	john1.jac	John	Jacobs	DATA220	A	95
8	tom.c@sj	Tom	C	DATA226	A	92
9	tom.c@sj	Tom	C	DATA220	B	89

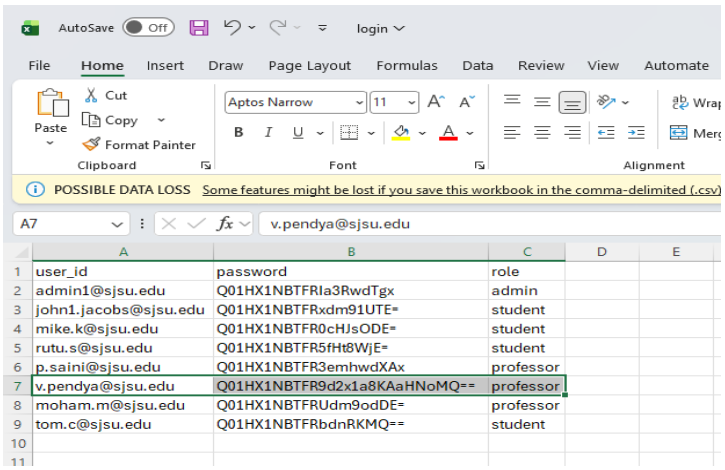
- Admin screen: User accounts: Reset a user's password
  - Before:login.csv file



	A	B	C
1	user_id	password	role
2	admin1@sjsu.edu	Q01HX1NBTFRIa3RwdTgx	admin
3	john1.jacobs@sjsu.edu	Q01HX1NBTFRxdm91UTE=	student
4	mike.k@sjsu.edu	Q01HX1NBTFR0cHJsODE=	student
5	rutu.s@sjsu.edu	Q01HX1NBTFR5fHt8WjE=	student
6	p.saini@sjsu.edu	Q01HX1NBTFR3emhwdXAx	professor
7	v.pendya@sjsu.edu	Q01HX1NBTFR9d2x1a8KAaDE=	professor
8	moham.m@sjsu.edu	Q01HX1NBTFRUdm9odDE=	professor
9	tom.c@sjsu.edu	Q01HX1NBTFRbdnRKMQ==	student

  - After: login.csv file for user v.pendyala@sjsu.edu

```
── User Accounts  [Admin] ──────────────────────────
1. View all accounts
2. Delete a user account
3. Reset a user's password
4. Change a user's role
5. System summary
0. Back
Select: 3
Email of user: v.pendyala@sjsu.edu
New password: vpendyala*
Error: User not found.

[Press Enter to continue]

── User Accounts  [Admin] ──────────────────────────
1. View all accounts
2. Delete a user account
3. Reset a user's password
4. Change a user's role
5. System summary
0. Back
Select: 3
Email of user: v.pendya@sjsu.edu
New password: vpendyala*
Password reset.
```



	A	B	C	D	E
1	user_id	password	role		
2	admin1@sjsu.edu	Q01HX1NBTFRIa3RwdTgx	admin		
3	john1.jacobs@sjsu.edu	Q01HX1NBTFRxdm91UTE=	student		
4	mike.k@sjsu.edu	Q01HX1NBTFR0cHJsODE=	student		
5	rutu.s@sjsu.edu	Q01HX1NBTFR5fHt8WjE=	student		
6	p.saini@sjsu.edu	Q01HX1NBTFR3emhwdXAx	professor		
7	v.pendya@sjsu.edu	Q01HX1NBTFR9d2x1a8KAaHNoMQ==	professor		
8	moham.m@sjsu.edu	Q01HX1NBTFRUdm9odDE=	professor		
9	tom.c@sjsu.edu	Q01HX1NBTFRbdnRKMQ==	student		
10					

- Admin screen: User accounts: System summary

```
── User Accounts  [Admin] ──────────────────────────
1. View all accounts
2. Delete a user account
3. Reset a user's password
4. Change a user's role
5. System summary
0. Back
Select: 5

── System Summary ──────────────────────────
Total students      : 8
Total courses       : 3
Total professors    : 3
Total user accounts : 8
──────────────────────────────────────────────
```

- Admin screen: Grade Reports & Statistics : Report by student

```
═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
 Logged in as: admin1@sjsu.edu  [admin]

 1. Student Management
 2. Course Management
 3. Professor Management
 4. User Accounts
 5. Grade Reports & Statistics
 6. Grade Scale Settings
 7. Change Password
 0. Logout
 Select: 5

 ── Grade Reports & Statistics  [Admin] ──────────────
 1. Report by student
 2. Report by course
 3. Report by professor
 4. Course average & median
 0. Back
 Select: 1
 Student email: rutu.s@sjsu.edu

 Grade Report — Student: rutu.s@sjsu.edu
 Email                    Name              Course     Grade   Marks
 ---------------------------------------------------------------------
 rutu.s@sjsu.edu          Rutu S            DATA226    B        89.0
 rutu.s@sjsu.edu          Rutu S            DATA200    A        96.0

 ─────────────────────────────────────────────────────────────────────
 Records : 2
 Average : 92.50    Median : 92.50

 [Press Enter to continue]
```

- Admin screen: Grade Reports & Statistics : Course average and median

```
[Press Enter to continue]

── Grade Reports & Statistics  [Admin] ──────────────
1. Report by student
2. Report by course
3. Report by professor
4. Course average & median
0. Back
Select: 4
Course ID: DATA200

Course  : DATA200
Average : 89.33
Median  : 86.00

[Press Enter to continue]
```

- Admin screen: Grade Scale Settings

```
═══════════════════════════════════════════════════════════
        CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════════
  Logged in as: admin1@sjsu.edu  [admin]

  1. Student Management
  2. Course Management
  3. Professor Management
  4. User Accounts
  5. Grade Reports & Statistics
  6. Grade Scale Settings
  7. Change Password
  0. Logout
  Select: 6

  — Grade Scale Settings ————————————————————————
  Grade       Min    Max
  ----------------------
  A          90.0  100.0
  B          80.0   90.0
  C          70.0   80.0
  D          60.0   70.0
  F           0.0   60.0

  1. Add grade level
  2. Modify grade level
  3. Delete grade level
  0. Back
  Select:
```

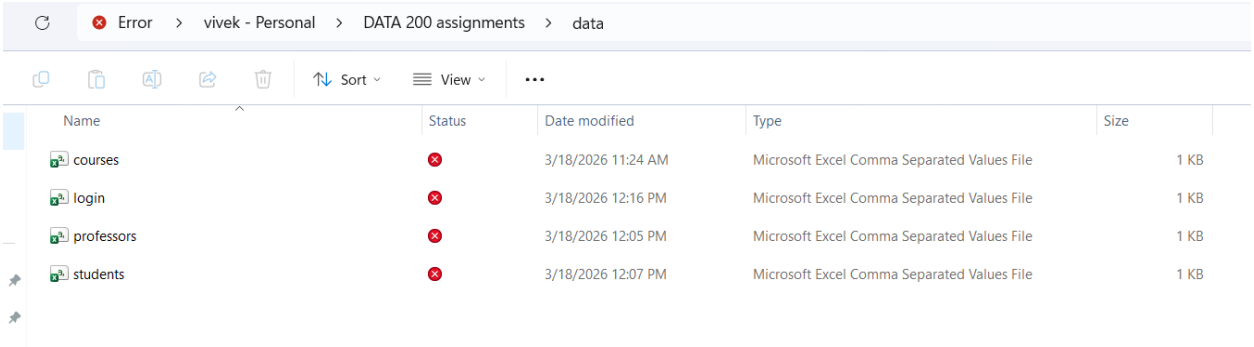
4. Login screen: Professor

```
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  1. Login
  2. Register
  0. Exit
  Select: 1

  — Login ————————————————————————————————————
  Email: p.saini@sjsu.edu
  Password: psaini*

  Welcome, p.saini@sjsu.edu! Role: professor

═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  Logged in as: p.saini@sjsu.edu  [professor]

  1. View my profile
  2. View my students
  3. Update a student's marks
  4. Grade report for my courses
  5. Course statistics
  6. Change my password
  0. Logout
  Select: 1
  Email   : p.saini@sjsu.edu
  Name    : Param Saini
  Rank    : Senior professor
  Course  : DATA200

  [Press Enter to continue]

═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  Logged in as: p.saini@sjsu.edu  [professor]

  1. View my profile
  2. View my students
  3. Update a student's marks
  4. Grade report for my courses
  5. Course statistics
  6. Change my password
  0. Logout
  Select: 2

  Email                    Name              Course     Grade   Marks
  ---------------------------------------------------------------------
  marc.j@sjsu.edu          Marc J            DATA200    B        86.0
  john1.jacobs@sjsu.edu    John Jacobs       DATA200    B        86.0
```

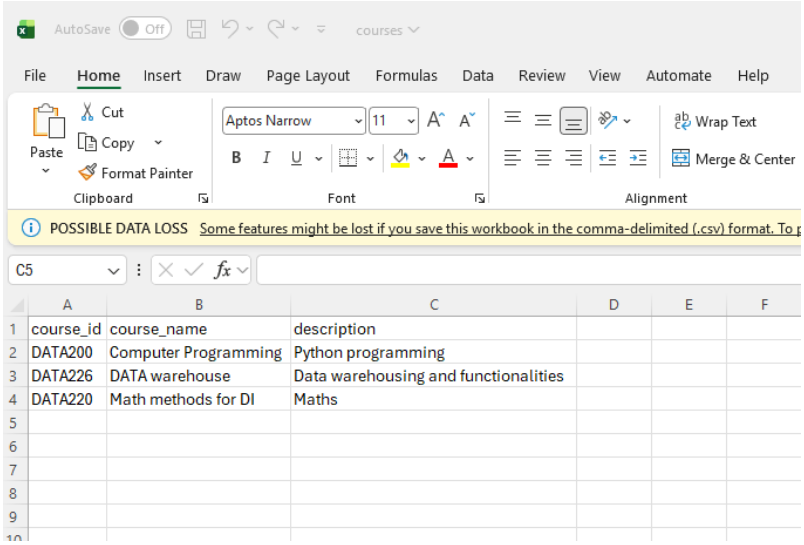
5. Login screen: Student

```
═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  1. Login
  2. Register
  0. Exit
  Select: 1

  ── Login ──────────────────────────────────────────
  Email: mike.k@sjsu.edu
  Password: mike1*

  Welcome, mike.k@sjsu.edu! Role: student

═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  Logged in as: mike.k@sjsu.edu  [student]

  1. View my profile & grades
  2. Enroll in another course
  3. Drop a course
  4. View available courses
  5. Change my password
  0. Logout
  Select: 1

  Grades for Mike K (mike.k@sjsu.edu)
  Course       Grade     Marks
  ------------------------------
  DATA220      B          85.0

  [Press Enter to continue]

═══════════════════════════════════════════════════════
      CheckMyGrade — SJSU Data Science Dept
═══════════════════════════════════════════════════════
  Logged in as: mike.k@sjsu.edu  [student]

  1. View my profile & grades
  2. Enroll in another course
  3. Drop a course
  4. View available courses
  5. Change my password
  0. Logout
  Select:
```

6. CSV files generated at the location

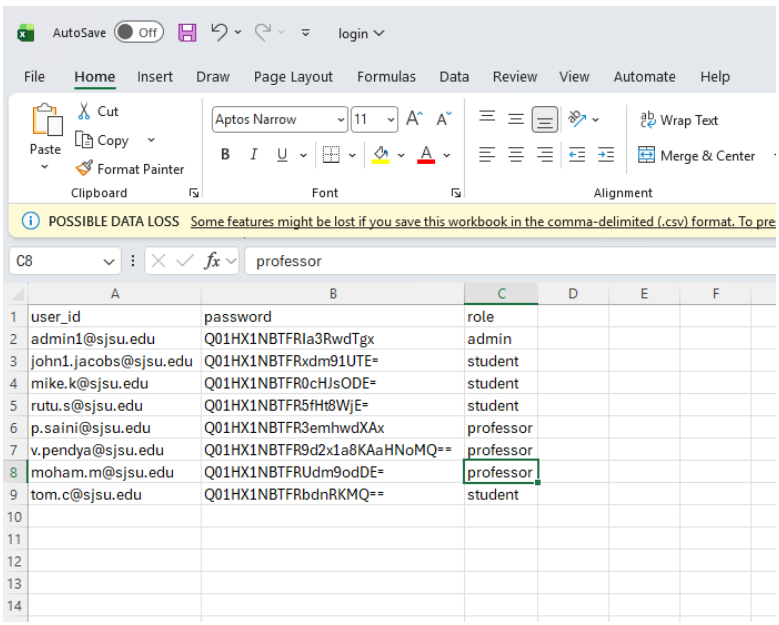


7. Course.csv file



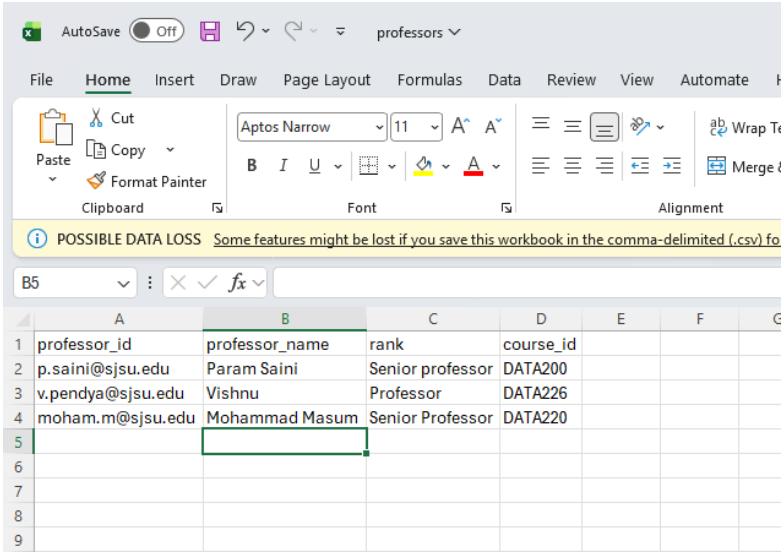
	A	B	C	D	E	F
1	course_id	course_name	description			
2	DATA200	Computer Programming	Python programming			
3	DATA226	DATA warehouse	Data warehousing and functionalities			
4	DATA220	Math methods for DI	Maths			
5						
6						
7						
8						
9						

8. Login.csv file – Password is in encrypted format



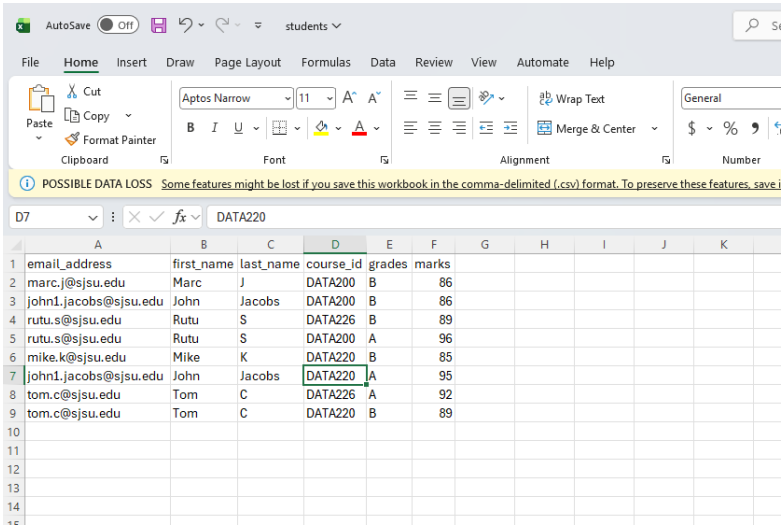
	A	B	C	D	E	F
1	user_id	password	role			
2	admin1@sjsu.edu	Q01HX1NBTFRIa3RwdTgx	admin			
3	john1.jacobs@sjsu.edu	Q01HX1NBTFRxdm91UTE=	student			
4	mike.k@sjsu.edu	Q01HX1NBTFR0cHJsODE=	student			
5	rutu.s@sjsu.edu	Q01HX1NBTFR5fHt8WjE=	student			
6	p.saini@sjsu.edu	Q01HX1NBTFR3emhwdXAx	professor			
7	v.pendya@sjsu.edu	Q01HX1NBTFR9d2x1a8KAaHNoMQ==	professor			
8	moham.m@sjsu.edu	Q01HX1NBTFRUdm9odDE=	professor			
9	tom.c@sjsu.edu	Q01HX1NBTFRbdnRKMQ==	student			
10						
11						
12						
13						
14						

9. Professors.csv file



professor_id	professor_name	rank	course_id
p.saini@sjsu.edu	Param Saini	Senior professor	DATA200
v.pendya@sjsu.edu	Vishnu	Professor	DATA226
moham.m@sjsu.edu	Mohammad Masum	Senior Professor	DATA220

10. Students.csv file

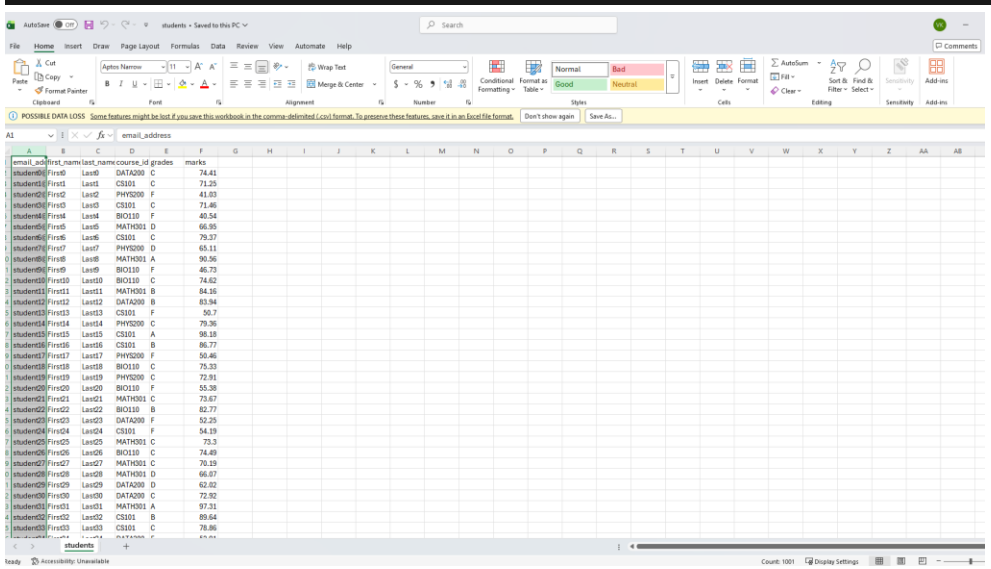


email_address	first_name	last_name	course_id	grades	marks
marc.j@sjsu.edu	Marc	J	DATA200	B	86
john1.jacobs@sjsu.edu	John	Jacobs	DATA200	B	86
rutu.s@sjsu.edu	Rutu	S	DATA226	B	89
rutu.s@sjsu.edu	Rutu	S	DATA200	A	96
mike.k@sjsu.edu	Mike	K	DATA220	B	85
john1.jacobs@sjsu.edu	John	Jacobs	DATA220	A	95
tom.c@sjsu.edu	Tom	C	DATA226	A	92
tom.c@sjsu.edu	Tom	C	DATA220	B	89

Unit Tests:

1. Testing of student records addition/deletion and modification. Perform the test to have the student files atleast 1000 records.

```
TERMINAL    PROBLEMS    OUTPUT    DEBUG CONSOLE    PORTS

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_zq4grfl3
test_admin_delete_user_cleans_profiles (__main__.TestRoleAndSync.test_admin_delete_user_cleans_profiles) ... ok
test_register_student_is_atomic (__main__.TestRoleAndSync.test_register_student_is_atomic) ... ok
test_role_change_blocks_inconsistent_profiles (__main__.TestRoleAndSync.test_role_change_blocks_inconsistent_profiles) ... ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_zq4grfl3

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_gqt773ev
test_binary_search_by_email (__main__.TestSearchAndSort.test_binary_search_by_email) ...
Binary search time: 0.0082 ms
ok
test_linear_search_by_course (__main__.TestSearchAndSort.test_linear_search_by_course) ...
Linear search time: 0.0753 ms
ok
test_sort_by_email (__main__.TestSearchAndSort.test_sort_by_email) ... ok
test_sort_by_marks (__main__.TestSearchAndSort.test_sort_by_marks) ...
Quick sort time: 0.8082 ms
ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_gqt773ev

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_i61t7gjp
test_add_delete_modify_student (__main__.TestStudentCRUD.test_add_delete_modify_student) ... ok
test_insert_1000_records_with_timing (__main__.TestStudentCRUD.test_insert_1000_records_with_timing) ...
Inserted 1000 student rows in 21845.46 ms

Inserted 1000 student rows in 21845.46 ms
Folder to inspect: C:\Users\vivek\AppData\Local\Temp\cmg_test_i61t7gjp
Press Enter after checking students.csv...
```



2. Load the data from previous runs saved in the csv files and you must be able to load the data and search.

```
Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_70depxus
test_binary_search_by_email (__main__.TestSearchAndSort.test_binary_search_by_email) ...
Binary search time: 0.0045 ms
ok
test_linear_search_by_course (__main__.TestSearchAndSort.test_linear_search_by_course) ...
Linear search time: 0.0687 ms
ok
```

3. You must have a test case to sort the student records (ascending/descending order) based on marks or student email_address. Your report must include the timing it

took to sort the records.

```
test_sort_by_email (__main__.TestSearchAndSort.test_sort_by_email) ... ok
test_sort_by_marks (__main__.TestSearchAndSort.test_sort_by_marks) ...
Quick sort time: 0.9201 ms
ok
Open folder in new window (ctrl + click)
```

4. Unit test to add/delete/modify the course

```
Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_dlztaj0r
test_add_delete_modify_student (__main__.TestStudentCRUD.test_add_delete_modify_student) ... ok
```

5. Unit test to add/delete/modify professors.

```
Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_y90z1jwy
test_add_delete_modify_professor (__main__.TestProfessorCRUD.test_add_delete_modify_professor) ... ok
test_professor_requires_existing_course (__main__.TestProfessorCRUD.test_professor_requires_existing_course) ... ok
```

```
PS C:\Users\vivek\OneDrive\DATA 200 assignments> & C:/Python314/python.exe c:/Users/vivek/Downloads/test_checkmygrade_aligned.py
Could not find platform independent libraries <prefix>

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_sekfgmjy
test_add_delete_modify_course (__main__.TestCourseCRUD.test_add_delete_modify_course) ... ok
test_prevent_delete_referenced_course (__main__.TestCourseCRUD.test_prevent_delete_referenced_course) ... ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_sekfgmjy

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_5_ojdue6
test_encrypted_differs (__main__.TestEncryption.test_encrypted_differs) ... ok
test_roundtrip (__main__.TestEncryption.test_roundtrip) ... ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_5_ojdue6

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_y90z1jwy
test_add_delete_modify_professor (__main__.TestProfessorCRUD.test_add_delete_modify_professor) ... ok
test_professor_requires_existing_course (__main__.TestProfessorCRUD.test_professor_requires_existing_course) ... ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_y90z1jwy

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_a0jf1lqp
test_admin_delete_user_cleans_profiles (__main__.TestRoleAndSync.test_admin_delete_user_cleans_profiles) ... ok
test_register_student_is_atomic (__main__.TestRoleAndSync.test_register_student_is_atomic) ... ok
test_role_change_blocks_inconsistent_profiles (__main__.TestRoleAndSync.test_role_change_blocks_inconsistent_profiles) ... ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_a0jf1lqp

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_70depxus
test_binary_search_by_email (__main__.TestSearchAndSort.test_binary_search_by_email) ...
Binary search time: 0.0045 ms
ok
test_linear_search_by_course (__main__.TestSearchAndSort.test_linear_search_by_course) ...
Linear search time: 0.0687 ms
ok
test_sort_by_email (__main__.TestSearchAndSort.test_sort_by_email) ... ok
test_sort_by_marks (__main__.TestSearchAndSort.test_sort_by_marks) ...
Quick sort time: 0.9201 ms
ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_70depxus

Created test folder: C:\Users\vivek\AppData\Local\Temp\cmg_test_dlztaj0r
test_add_delete_modify_student (__main__.TestStudentCRUD.test_add_delete_modify_student) ... ok
test_insert_1000_records_with_timing (__main__.TestStudentCRUD.test_insert_1000_records_with_timing) ...
Inserted 1000 student rows in 21819.49 ms
ok
test_requires_existing_course (__main__.TestStudentCRUD.test_requires_existing_course) ... ok
test_same_student_multiple_courses (__main__.TestStudentCRUD.test_same_student_multiple_courses) ... ok

Test folder kept at: C:\Users\vivek\AppData\Local\Temp\cmg_test_dlztaj0r

----------------------------------------------------------------------
Ran 17 tests in 42.026s

OK
```